Year 10 Higher

REVISION PACK

Graphs 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 1 · Gradients and equations of lines (Lessons 3.1, 3.3)

Q	Working	Answer
a missing coordinate		
1.1	$\frac{k-2}{7-3}=4$ $k-2=16$ $k=18$	$k=18$
1.2	$\frac{k-9}{4-1}=-2$ $k-9=-6$ $k=3$	$k=3$
1.3	$\frac{k-5}{3+2}=3$ $k-5=15$ $k=20$	$k=20$
1.4	$\frac{k--3}{4-0}=2$ $k--3=8$ $k=5$	$k=5$
1.5	$\frac{k-1}{5-2}=-3$ $k-1=-9$ $k=-8$	$k=-8$
1.6	$\frac{k-6}{3+1}=2$ $k-6=8$ $k=14$	$k=14$
the equation of a line through two points		
1.7	$m=\frac{12}{3}=4$ $7=4\times(2)+c$ $c=-1$	$y=4x-1$

1.8	$m = \frac{12}{4} = 3$ $-1 = 3 \times (-1) + c$ $c = 2$	$y = 3x + 2$
1.9	$m = \frac{-9}{3} = -3$ $7 = -3 \times (-1) + c$ $c = 4$	$y = -3x + 4$
1.10	$m = \frac{-12}{3} = -4$ $6 = -4 \times (1) + c$ $c = 10$	$y = -4x + 10$
1.11	$m = \frac{-8}{4} = -2$ $10 = -2 \times (-3) + c$ $c = 4$	$y = -2x + 4$
1.12	$m = \frac{10}{2} = 5$ $-4 = 5 \times (0) + c$ $c = -4$	$y = 5x - 4$

Graphs 1 · Drawing straight-line graphs (Lesson 3.2)

Q	Working	Answer
a table of values and a graph		
2.1	y values: $-7, -4, -1, 2, 5, 8$	**Straight line through** $(-2, -7)$ **and** $(3, 8)$
2.2	y values: $6, 4, 2, 0, -2, -4$	**Straight line through** $(-1, 6)$ **and** $(4, -4)$
2.3	y values: $1, 1.5, 2, 2.5, 3, 3.5, 4$	**Straight line through** $(-2, 1)$ **and** $(4, 4)$
2.4	y values: $5, 4, 3, 2, 1, 0, -1$	**Straight line through** $(-1, 5)$ **and** $(5, -1)$